Chapter 2, Problem 60P

Problem

If the three bulbs of Prob. 2.59 are connected in parallel to the 120-V source, calculate the current through each bulb.

Step-by-step solution

Step 1 of 2

Three bulbs of $40\ \text{W}$, $60\ \text{W}$ and $100\ \text{W}$ are connected in parallel to $120\ \text{V}$ source. The circuit is as shown in Figure 1.

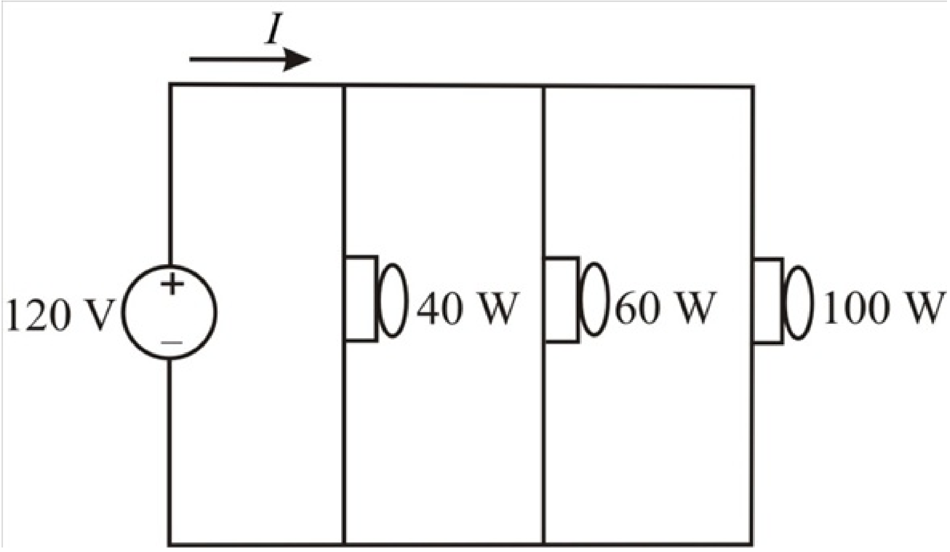

Figure 1

Step 2 of 2

The current through $40\ \text{W}$ bulb is,

$$\begin{aligned} I_{40\text{W}} &= \frac{P_{40\text{W}}}{120} \\ &= \frac{40}{120} \\ &= 0.33\ \text{A} \end{aligned}$$

The current through $60\ \text{W}$ bulb is,

$$\begin{aligned} I_{60\text{W}} &= \frac{P_{60\text{W}}}{120} \\ &= \frac{60}{120} \\ &= 0.5\ \text{A} \end{aligned}$$

The current through $100\ \text{W}$ bulb is,

$$\begin{aligned} I_{100\text{W}} &= \frac{P_{100\text{W}}}{120} \\ &= \frac{100}{120} \\ &= 0.833\ \text{A} \end{aligned}$$

Therefore, the currents through the bulbs are $\boxed{I_{40\text{W}} = 0.33\text{A}, I_{60\text{W}} = 0.5\text{A}, I_{100\text{W}} = 0.833\text{A}}$.